

Radical equations



1

a $x = 13$

b $h = 25$

c $x = \frac{9}{64}$

d $k = 16$

2 A radical equation is an equation where the variable is inside a radical.

3 No

4 No

5 Yes

6

a $y = 49$

b $x = 25$

c $p = 9$

d $y = \frac{9}{25}$

e $x = \frac{49}{144}$

f $x = 64$

g $y = 4$

h $c = 9$

i $b = 25$

j $a = 9$

k $b = \frac{1}{25}$

l $a = 4$

7

a $x = 30$

b $x = 5$

c $x = 54$

d $x = 11, -3$

e $x = 4, 2$

f $x = 18$

g $x = 16$

h $x = 22$

i $x = 135$

j $x = 121$

k $x = 4, -4$

8

a No

b Ray did not know that $\sqrt{x+3}$ must always be non-negative. So at step 2, he should have known there was no solution.

9

a Tom did not divide every term by 2 in step 2, so he should have also divided -10 by 2.

b In step 4 Katrina took the square root of 9, when she should have squared it.

c $x = 10$

10 Example answers:



- In order for the left-hand side to equal zero, the square root of the number would have to be negative.
- The left-hand side is greater than zero for all values of x .

11

a $x = \frac{1}{4}, \frac{1}{25}$

b $x = \frac{1}{25}$

12

a $x = 9$

b No

13 Raise each side of the equation to the power $\frac{5}{3}$.

14

a $x = 57$

b $x = 4$

c $x = 79$

15

a $x = 9$

b $x = \pm 128$

c $x = 81$

d $x = \pm 128$

Applications

16

a

V	40	400	4 000
b	3.2	10.0	31.6

b No

17

a $t = \sqrt{4-x} + 4$

b $t = 6$

c 0.2 seconds

18

a 8

b Fresh gale

c $s = 115$

d $55 \leq s \leq 64$